A Successful Story of Applying a Variety of Representational Tools in College STEM Education [1]

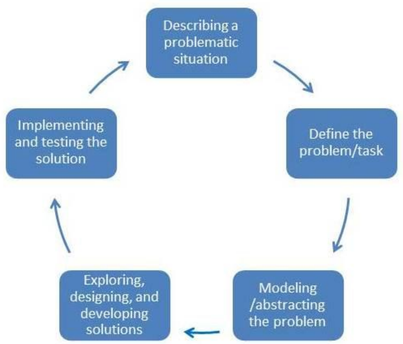
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Background

The STEM Education Center is working with faculty in the Electrical and Computer Engineering Department to adapt the Lesh Translational Model, though specifically designed for elementary mathematics teachers in an electrical engineering classroom. The current study reports a successful story of using a variety of representational tools for instruction.

Iterative cycles of engineering practices



Lesh Translational Model

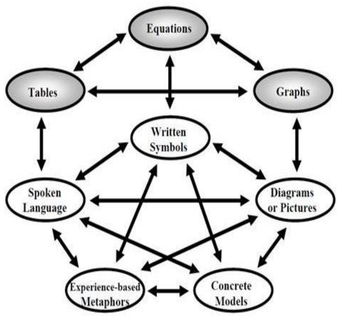


Fig. 1 (L) Engineering Practice Cycles; (R) Lesh Translational Model.

Research Questions

(1) Is content knowledge co-developed with problem-solving skills and metacognitive functions under problem-centered learning?

(2) Does the distribution of content knowledge among a variety of representational modes facilitate such co-development?

(3) What are the implications in STEM teaching and learning for such development?



Reference:
Lesh & Doerr, "Foundations of a Models and Modeling Perspective on Mathematics", in *Beyond Constructivism: Models and modeling perspectives on mathematics,* (Lesh and Doerr, Eds). Mahwah, NJ: Lawrence Erlbaum Associates; (2003), pp 3-34.

Purpose

- To expand on innovative practices for "flipping lectures" into recitations;
- To prepare students to face real world challenges focusing on co-development of cognitive and metacognitive knowledge and skills.

Research Design and Method

Applying Design-Based-Research (DBR) to intertwine the three goals of research, design, and pedagogical practices in this study.

Data collection from multiple courses including:

- two on-line surveys at the beginning and the end of the semester
- students' exam papers and other worksheets

Data analyses:

- Developing a framework to evaluate the co-development
- Developing scoring rubrics to assess representational fluency

Results and Findings

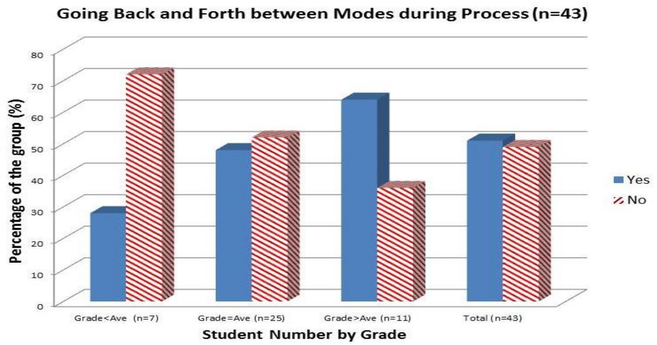


Fig. 2 Percentage of the student groups that shifted usages of representational modes.

[1] *Lin & Imbertson, Conference Proceedings, 122th ASEE Annual Conference and Exposition, 2015; References therein.*

Overall Score	Diagram (DI)		Equation (EQ)		Student Number (n)
	Sub-score	Student number(n1)	Sub-score	Student number (n2)	
0					2
0.5	0.5	1	0.5	5	6
1	0.5	5	0.5	5	5
1.5	0.5	2	1.0	13	15
2	1.0	13	1.0	13	13

Student number \ Mode applied	Label (LA)	Equation (EQ)	Text (TX)	Diagram (DI)	NA
Using specific mode	17	5	4	16	6
Using additional mode(s)	1	3	3	3	

Fig. 3 Improved translational fluency (upper: test 1; lower: final).

Metacognitive for Cognitive Activities in Problem Solving

Metacognitive Challenges	Cognitive Activities in Problem Solving					
		Focus	Define	Devise	Execute	Evaluation
	Interpreting meanings of the problem	Important information and features	Essential concepts	Strategies & Approach		
		Structure	Target quantity & Goals	Knowledge organization and applicability		
	Monitoring & Self-regulation				Following through Tracking the progress	
	Reflection			Procedural efficiency		Quality

Findings (Figs 2&3)

- Students improved abilities of using multiple representational tools;
- The Level of improvement was positively correlated with course performance .

Conclusion and Implications

Multiple representations embedded in instructions

- ❑ promote teaching and learning that facilitates the co-development of content knowledge, problem solving skills, and metacognitive knowledge;
- ❑ strengthen the association of instructional objectives for lectures and recitations;
- ❑ require that problems are framed in ways to engage students in deep discussions during lecture periods and provide students with resources and insights in problem solving during recitations;
- ❑ offer opportunities for formative assessment allowing instructors to understand what and how students learn.